Hedi Tebourbi

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Bachelor in Computer Science (BiCS)
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Professional Experience

- **FSTM, Department of Computer Science – University of Luxembourg** *April 2026 – Present*
  Research Assistant Luxembourg
  - –**Main focus:** developing ATLAS, the EIF-funded Socratic AI tutoring platform (see Projects): a full-stack FastAPI / Next.js multi-agent system with RAG grounded in educator course materials and a teacher dashboard for learning analytics.
  - –Alongside, research on LLM-based annotation and retrieval-augmented evaluation for Luxembourgish, a low-resource language, building reproducible Python pipelines that orchestrate OpenAI, Anthropic, and DeepSeek APIs across few-shot, zero-shot, and chain-of-thought paradigms, with self-hosted Langfuse observability (Docker) for full prompt traceability and token/cost tracking.
  - –Designed evaluation frameworks scoring LLM and fine-tuned-encoder systems against expert human annotations across 11 diagnostic linguistic categories, using leakage-free cross-validation, unified precision/recall/F1 metrics, and structured per-run JSONL reporting.
  - –Joint first author of two *LaTeLL 2026* papers, LuxDiag-RAG and a human-in-the-loop multi-LLM study on expanding Luxembourgish educational data (see Publications); also delivered a fine-tuned-encoder vs. multi-LLM annotation comparison as a reproducible open-source study (see Projects).

- **LISER – Luxembourg Institute of Socio-Economic Research** *March 2026 – June 2026*
  Research Assistant Luxembourg
  - –Architected the MIGLINK University Harmonization pipeline, mapping 594M LinkedIn education records to ROR-standardized university identities across 63M+ unique raw institution strings, running at scale on the MeluXina supercomputer (SLURM).
  - –Designed a cascading matching system (exact match → suffix-normalized match → LLM match) achieving **100% precision** and **94.8% recall** on an annotated ground truth; integrated Qwen3-32B LLM inference via vLLM on GPU cluster with a hard validation gate against the Research Organization Registry (ROR v2.4) to eliminate hallucinated matches.
  - –Built a multilingual institution type classifier with Unicode script detection across 9 writing systems (Latin, CJK, Arabic, Cyrillic, Devanagari, Korean, Thai, Greek, Hebrew) and a keyword vocabulary spanning 12+ languages, augmented by a curated override dictionary for brand-name disambiguation; achieves 100% university-match precision on ground truth.
  - –Engineered the full data pipeline in Python (Polars streaming, 18 scripts) with automated formatting, a reusable ground truth evaluation harness, and mandatory per-record traceability fields covering match tier, confidence score, and harmonization source.

- **AI Robolab - University of Luxembourg** *September 2024 – March 2026*
  Research Assistant Luxembourg
  - –Led and coordinated the development of the Luxembourgish Language Learning App with a student team, managing the GitHub repository, reviewing pull requests, assigning tasks, and ensuring project progress.
  - –Enhanced and maintained the backend by resolving legacy issues, introducing robust error handling, simplifying installation, and addressing dependency conflicts for new features such as TTS/STT integration.
  - –Contributed to feature development including an automated image cropping tool and dynamic image retrieval via function calling.
  - –Continued the development of the AIA-PAL project, focusing on backend improvements and adaptive content generation.
  - –Designed and implemented a benchmarking pipeline for open-source LLMs, conducting extensive evaluation to identify cost-efficient models capable of replacing ChatGPT while maintaining strong performance in Luxembourgish.
  - –Co-authored and published multiple peer-reviewed papers (see Publications).
  - –Managed the university's AI Robolab, organizing and delivering demonstrations of social robots (e.g., SoftBank Pepper, Furhat, LuxAI QTrobot), engaging students and visitors during lab sessions and public outreach events.
  - –Represented the laboratory as part of a research team at the CALM '25 conference in Luxembourg, showcasing ongoing projects and the interactive capabilities of Furhat and QTrobot to the academic community.

- **Deelan.ai** *March 2025 – June 2025*
  Junior Software Developer (Startup) Luxembourg
  - –Contributed to the development of an AI-powered training platform for SMBs, enabling rapid generation of structured courses from prompts or uploaded documents.

– Worked on onboarding and authentication flows, improving UX, navigation, and reliability across signup, login, and freemium trial access.
– Implemented and refined a freemium trial system, including role-based access control, trial expiration logic, email verification, and user feedback.
– Integrated website analysis and scraping into onboarding to extract structured company information for course generation.
– Contributed to database schema evolution related to user roles and metadata, ensuring correct role handling during authentication.
– Delivered a client-facing WordPress CMS migration (Webflow → WordPress), designing custom post types, SEO-oriented content schemas, and dynamic metadata injection.
– Built reusable Elementor templates and a no-code editorial workflow, enabling scalable content creation and long-term SEO maintainability.

Projects

- **ATLAS – Adaptive Tutoring & Learning via AI Scaffolding** 🔗 *September 2025 – Present*
  AI-powered educational platform | EIF-funded initiative, University of Luxembourg
  – Designed and developed a curriculum-agnostic AI tutoring platform that employs Socratic dialogue to guide learners through questioning rather than direct answers, countering the "cognitive debt" problem identified in recent AI-in-education research.
  – Built a full-stack architecture with a FastAPI backend and Next.js frontend, featuring a hybrid multi-agent system (CrewAI) that separates real-time tutoring from background analysis to maintain 2–4s response latency.
  – Implemented RAG grounded in educator-provided course materials, adaptive onboarding for learner profiling, and a teacher dashboard surfacing real-time learning analytics and misconception tracking.
  – Co-authored and submitted a grant application to the Education Innovation Fund (EIF/I^{2}TL) alongside Prof. Leon Van Der Torre (PI) and Dr. Salima Lamsiyah, covering all narrative sections and budget justification.
  – Presented the project at the I^{2}TL Teaching Day to positive feedback, sparking interest in a cross-disciplinary pilot in medicine; also featured in a university article 🔗 by the FSTM Communication Office.

- **Prosody Manipulation for Voice-Based AI Agents** 🔗 *September 2025 – Present*
  Bachelor Semester Project 5 & 6 (BSP)
  – Research project investigating the role of prosody manipulation in voice-based AI systems, with a focus on persuasion, engagement, and user perception in banking and insurance interaction scenarios.
  – Designed experimental pipeline with Text-to-Speech (TTS) models to control prosodic features such as pitch, tempo, emphasis, and emotional tone.
  – Contributed to the definition of experimental protocols, scenario design, and stimulus generation for human-subject studies.
  – Conducted qualitative interviews with study participants to gather subjective feedback on voice agent perception and decision-making.
  – Organized and coordinated the user study, including participant recruitment, session scheduling, and on-site facilitation.
  – Built a full-stack web application for participant data collection (React, Node.js, Express, SQLite), featuring a phone call simulation interface, randomised stimulus presentation, and a real-time researcher dashboard with live monitoring via WebSockets.
  – Co-authored a paper accepted at *CUI 2026*, presenting findings on prosodic influence in voice-based financial AI scenarios.

- **LuxDiag Workbench: Interactive Diagnostic Interface** 🔗 *June 2026 – July 2026*
  DCS, University of Luxembourg
  – Built an interactive, browser-based diagnostic workbench that unifies my two prior Luxembourgish studies, a diagnostic RAG evaluation (LuxDiag-RAG) and an encoder-vs-LLM annotation study, on a shared character-offset coordinate system, so retrieved chunks, evidence spans, model answers, and annotated tokens all index the same `[start, end)` offsets in one canonical text and can be inspected together one example at a time.
  – Designed four linked panels (retrieval, answering, retrieval-trap explorer, and annotation) driven by a single URL-encoded selection state, with clickable cross-layer and cross-text span highlighting, a guided tour, an in-place glossary, and shareable, reproducible views.
  – Engineered a data-driven architecture in which the entire interface is a pure function of a JSON schema, so any conforming study, including a contrasting English example bundle and user-uploaded datasets, re-renders in the browser with no code changes; added a keyless live retriever (character-BM25, no API keys or LLM calls).
  – Developed the front end with Vite, React 19, Tailwind CSS v4, and Recharts, and deployed it on Vercel as a static SPA plus one serverless function (live demo 🔗).
  – Co-authored a system-demonstration paper submitted to *EMNLP 2026 System Demonstrations*, accompanied by a screencast 🔗 and a public live demo.

- **Fine-Tuned Encoders vs. LLMs for Luxembourgish Diagnostic Annotation** 🔗 *April 2026 – June 2026*
  DCS, University of Luxembourg

– Compared LoRA fine-tuned encoder models (LuxemBERT, XLM-RoBERTa, ModernBERT) against a multi-LLM annotation pipeline for eleven diagnostic linguistic categories on the LuxDiagRC Luxembourgish reading-comprehension corpus.
– Demonstrated that fine-tuned encoders outperform the LLM pipeline (LuxemBERT 38.7% macro-F1 / 78.0% micro-F1 vs. 30.7% / 24.2% for the best LLM), with all training running on a single 4 GB consumer GPU (GTX 1050 Ti) in under two hours.
– Built the token-classification dataset, a leakage-free 4-fold cross-validation, and a shared precision/recall/F1 evaluation; uncovered a three-regime division of labour where encoders win surface morpho-orthographic categories and LLMs retain world-knowledge categories.
– Tested ensemble pseudo-labelling as data augmentation, establishing a negative result that the world-knowledge gap is a knowledge limitation rather than a data-quantity one.
– Published the complete pipeline and experiments as a reproducible open-source repository.

- **LuxDiag-RAG: Diagnostic RAG Evaluation for Luxembourgish Reading Comprehension** *May 2026 – June 2026*
  DCS, University of Luxembourg

  – Built a diagnostic evaluation of retrieval-augmented generation (RAG) for Luxembourgish multiple-choice reading comprehension on the LuxDiagRC corpus (640 questions over 16 annotated texts), separating retrieval failure from comprehension failure across closed-book, full-text oracle, text-restricted RAG, and open-corpus RAG settings.
  – Compared lexical (word- and character-level BM25), multilingual dense (multilingual-E5, BGE-M3), and hybrid retrievers, showing that character n-gram BM25 and weighted hybrid fusion beat off-the-shelf multilingual dense retrieval for open-corpus evidence selection, driven by a measured orthography gap between informal questions and literary source texts.
  – Evaluated four frontier LLMs (GPT-5.5, DeepSeek v4-pro, Claude Opus 4.8, Sonnet 4.6) with span-level evidence-recall metrics and a retrieval-trap analysis quantifying how retrieving a distractor span without the critical span inflates the rate of choosing the lure.
  – Shipped a reproducible Python pipeline (data, retrieval, answering, analysis) with 197 automated tests, and co-first-authored a paper published at *LaTeLL 2026* (Fes, Morocco).

- **KlangWee – AI Concert Journey Composer** *May 2026 – June 2026*
  Student AI Lab Challenge – Philharmonie Luxembourg

  – Built a full-stack AI companion for the Philharmonie Luxembourg season that curates personalised three-concert journeys and reveals a hidden musical arc after the user attends all three concerts.
  – Designed a multi-agent LangGraph pipeline with a Knowledge Graph-driven concert selector, a streaming narrator agent, and a critic node; arcs are scored by a dramaturg formula weighting user alignment, connection strength, and discovery.
  – Engineered a slot-fill fallback using DeepSeek to bypass the Knowledge Graph when pin constraints cannot be satisfied, querying the full catalogue and selecting replacements via LLM.
  – Built the backend in FastAPI (async SQLAlchemy, PostgreSQL) and the frontend in Nuxt 3 / Vue 3 with SSE streaming, Pinia state management, and a Philharmonie editorial design system (sharp corners, teal accent, Helvetica Neue).
  – Integrated Spotify's Client Credentials API for per-concert artist embed previews and implemented a multi-select check-in form feeding into the hidden-arc reveal mechanic.
  – Deployed with Docker Compose; includes demo mode, admin analytics dashboard, and a shareable journey token system.

- **T-MAP: A Transition-aware Framework for Collaborative Spatial Mind-Mapping in VR** *September 2025 – May 2026*
  User Centered Design course project

  – A transition-aware framework organising collaborative VR mind-mapping into five layers: ideation, structuring, collaboration, awareness, and convergence, addressing the transition cost of moving between private ideation, shared editing, review, and consensus.
  – Defined a quantitative transition burden index (z-scored composite of transition duration, help requests, edit conflicts, and repair utterances) and four hypotheses (H1–H4) covering transition continuity, shared awareness/role clarity, coordination/convergence, and ideational fluency as a negative control.
  – Contributed to the design and execution of a within-subjects pilot study with 10 two-person teams (20 participants) comparing a baseline shared VR map against the T-MAP version on Meta Quest standalone headsets, using Wilcoxon signed-rank tests with Holm correction across subjective, behavioural, and map-coding outcome families.
  – Instantiated transition-support mechanisms in a Unity node-link prototype, including private/shared idea states, role and focus indicators, review states, consensus pins, and snapshot/rollback recovery.
  – Findings showed T-MAP reduced transition burden (B: 0.74 → -0.74) and improved flow continuity, shared awareness, role clarity, and map coherence, without increasing the raw number of ideas generated, supporting the framework's core claim that making idea, role, review, and agreement states explicit reduces collaboration burden.
  – Paper submitted to *XRTransitions'26*, the First International Workshop on the Multiple Dimensions of the Transitions in Collaborative eXtended Reality.

- **Coffee Shop App** *September 2025 – February 2026*
  Software Engineering course project

  –Team-based software engineering project focused on the design and development of a coffee shop management application.
  –Contributed to requirements analysis, system design, and implementation following software engineering best practices.
  –Implemented core application features, collaborated using Git for version control, and participated in code reviews and task coordination.
  –Applied principles of modular design, maintainability, and documentation throughout the development process.

- **LetzLearn – AI-Powered Luxembourgish Language Learning App** *September 2025 – January 2026*
  Generative AI and LLM-based Agents course project

  –Designed and developed an AI-powered web application for learning Luxembourgish through interactive conversations, quizzes, vocabulary training, and speech practice.
  –Implemented multiple learning modes (conversation, grammar, vocabulary, quiz, immersion) using an LLM with real-time streaming responses and RAG for course-aware answers.
  –Built a spaced-repetition vocabulary system (SM-2 inspired) with mastery tracking, daily challenges, and detailed progress analytics.
  –Integrated speech technologies including text-to-speech (ElevenLabs) and speech-to-text for pronunciation and dialog practice.
  –Developed a full-stack architecture using Next.js, React, Supabase (PostgreSQL + Auth + RLS), and Vercel AI SDK.
  –Deployed and tested the system with persistent user profiles, conversation history, progress dashboards, and secure data access.

- **AI-Powered Luxembourgish Language Learning App** *September 2024 – December 2025*
  AI Robolab Project

  –Responsible for maintaining, updating, and enhancing an AI-powered web application for learning Luxembourgish through interactive, intelligent agents built with LangGraph.
  –Designed and implemented a database for efficient management of user data, profile synchronization, and personalized learning content.
  –Developed frontend features, resolved bugs, and continuously improved the React-based user interface for optimal user experience.
  –Actively contributed to ongoing project improvements, backend optimizations, and integration of user feedback.
  –This work resulted in peer-reviewed publications and ongoing submissions (see Publications).

- **Artificial Intelligent Agents for Personalized Adaptive Learning (AIA-PAL)** *September 2024 – June 2025*
  Bachelor Semester Project 3 (BSP)

  –Initiated as part of my BSP3 and further developed during my employment as a research assistant.
  –Developed an adaptive learning backend leveraging LangGraph and CrewAI within a Multi-Agent System (MAS) to dynamically tailor educational content based on student interactions and learning pace.
  –Designed AI agents capable of adjusting course material and quizzes in real time according to student performance and chatbot interactions.
  –Built entirely in Python using LangGraph and its dedicated IDE, LangGraph Studio; actively contributed to ongoing research and development.
  –This project resulted in a peer-reviewed publication and presentation (see Publications).

- **Personalized Learning with ChatGPT** *February 2024 – July 2024*
  Bachelor Semester Project 2 (BSP)

  –Developed a personalized learning platform powered by CrewAI and ChatGPT API to assist students in studying efficiently.
  –Implemented features allowing students to upload course PDFs and receive tailored content, including customized course material, quizzes, and flashcards, based on their learning preferences (e.g., more examples, detailed explanations, or specific styles).
  –Integrated an AI chatbot capable of answering course-related questions, providing feedback, and assisting with quizzes.
  –Designed and developed the backend using Django and CrewAI, and built an interactive frontend using React.

- **HomeLab Server (Self-Hosted Infrastructure)** *Personal Project*
  Personal homelab built from repurposed hardware for self-hosting and experimentation

  –Converted an old computer into a dedicated homelab server running Linux, used as a sandbox for self-hosted services and infrastructure experimentation.
  –Deployed AdGuard Home for network-wide DNS filtering and privacy-aware ad blocking.

–Managed the system using Cockpit for server administration and monitoring.
–Used Docker extensively, with Portainer for container orchestration and lifecycle management.
–Configured NGINX Proxy Manager for reverse proxying and secure service exposure.
–Set up File Browser to manage storage, local cloud functionality, and file access.
–Self-hosted applications including n8n, open-source AI tools, and other services for automation and experimentation.
–Used SSH for secure remote access, server management, and automation workflows, integrated with Tailscale to enable secure remote access to the homelab over a private mesh VPN.
–Developed practical expertise in Docker, Linux system administration, networking, VPNs, reverse proxies, and self-hosted application architectures.

Publications

- **AAMAS '25: 24th International Conference on Autonomous Agents and Multiagent Systems**
  Personalized Language Learning: A Multi-Agent System Leveraging LLMs for Teaching Luxembourgish.
  H. Tebourbi, S. Nouzri, Y. Mualla, A. Najjar. [PDF]
- **MobiSPC '25: 22nd International Conference on Mobile Systems and Pervasive Computing**
  Artificial Intelligence Agents for Personalized Adaptive Learning.
  H. Tebourbi, S. Nouzri, Y. Mualla, A. Abbas-Turki. [DOI]
- **MDPI Applied Sciences, July 2025.**
  BPMN-Based Design of Multi-Agent Systems: Personalized Language Learning Workflow Automation.
  H. Tebourbi, S. Nouzri, Y. Mualla, M. El Fatimi, A. Najjar, A. Abbas-Turki, M. Dridi. [DOI]
- **EUSPN '25: 16th International Conference on Emerging Ubiquitous Systems and Pervasive Networks**
  Can Visual Attention Serve as an Explanation?
  P. Kluczynski, F.-Z. El-Qoraychy, Y. Mualla, M. Dridi, H. Tebourbi, S. Nouzri. [DOI]
- **CALM '25: 2nd International Workshop on Causality, Agents and Large Models**
  Augmenting an LLM-Based Tutor Agent with a Large Action Model for Multimodal Interaction.
  P. Kluczyński, Y. Mualla, S. Nouzri, H. Tebourbi, A. Picard, F. Gechter, A. Abbas-Turki. [DOI]
- **ACM CUI '26: 8th ACM International Conference on Conversational User Interfaces**
  How Authoritative Can Synthetic Voices Sound?
  Prosodic Manipulation and Perceptual Evaluation Across Two TTS Platforms.
  H. Tebourbi, A. Stassart, S. Rastegar, S. Vijapurapu, A. Sergeeva, A. Leschanowsky. [DOI]
- **LaTeLL '26: International Conference on Language Technologies for Low-resource Languages**
  LuxDiag-RAG: Diagnostic Evaluation of Retrieval-Augmented Generation
  for Luxembourgish Reading Comprehension.
  K. Murugaraj*, H. Tebourbi*, C. Friezas Gonçalves, S. Lamsiyah. *(*equal contribution)* [PDF]
- **LaTeLL '26: International Conference on Language Technologies for Low-resource Languages**
  Assessment of Human-in-the-Loop Multi-LLMs for Low-Resource Educational Data Expansion.
  C. Friezas Gonçalves*, H. Tebourbi*, C. Schommer, S. Lamsiyah. *(*equal contribution)* [PDF]

Academic Service & Peer Review

•Invited as reviewer for international conferences and workshops in Artificial Intelligence, Human–Computer Interaction, and Multi-Agent Systems, and member of the Technical Program Committee of CALM 2027.

- **The 3rd International Conference on Artificial Intelligence and Human-Computer Interaction**
  (ArtInHCI 2025), Nanning, China, Oct 2025
- **The 2nd International Workshop on Causality, Agents and Large Models**
  (CALM 2025), Luxembourg, Luxembourg, Dec 2025
- **The 3rd International Workshop on Causality, Agents and Large Models**
  (CALM 2026), Istanbul, Türkiye, Apr 2026
- **The 25th International Conference on Autonomous Agents and Multiagent Systems**
  (AAMAS 2026), Demonstrations Track, Paphos, Cyprus, May 2026
- **The 23rd International Conference on Mobile Systems and Pervasive Computing**
  (MobiSPC 2026), Athens, Greece, Aug 2026
- **The 17th International Conference on Emerging Ubiquitous Systems and Pervasive Networks**
  (EUSPN-2026), Almaty, Kazakhstan, Oct 2026
- **The 4th International Workshop on Causality, Agents and Large Models – Technical Program Committee**
  (CALM 2027), Delft, the Netherlands, Apr 2027

Extracurricular Activities

- **Association for Computing Machinery (ACM)** Student Member *(Member No. 8409478)* *2026 – Present*
- **Student Association for University Campus Life (SAUL)** *September 2024 – Present*
  Board Member – Social Media Manager

  –Elected board member responsible for managing and developing the association's social media strategy and online presence.
  –Designed and published digital content to promote campus events, increase student engagement, and strengthen brand visibility.
  –Coordinated event communication campaigns and collaborated with other board members to align outreach with organizational objectives.
  –Acted as shift responsible during campus events and parties, overseeing setup, bar service, team coordination, and post-event logistics to ensure smooth operations.

- **Study Programme Representative (SPR)** *2023 – 2026*
  Bachelor in Computer Science

  –Represented student interests and concerns impartially, collecting requests, complaints, and suggestions and presenting them to the appropriate committees or responsible persons.
  –Met regularly with the Course Director and acted as a central point of contact for student-related academic matters.
  –Represented students in the Steering / Advisory Committee (*Comité de pilotage*) of the study programme in accordance with Article 70 of the Study Regulations.

- **Lex Vitae Podcast** *October 2024 – February 2025*
  Co-Host & Co-Creator

  –Co-created and co-hosted a podcast aimed at helping university students achieve academic success and maintain life balance through evidence-based tips and practical strategies.
  –Produced episodes on critical student topics, including *Essentials of Sleep* and *From Stress to Self-Care: Building Emotional Well-Being in University* featuring expert psychologist guest.
  –Authored and launched a companion newsletter on Substack, providing in-depth resources, practical guidance, and curated scientific papers related to episode themes.
  –Expanded audience reach by publishing on major platforms such as Spotify 🔗, Amazon Music 🔗, and YouTube 🔗, incorporating video content for enhanced engagement.

International Academic Experience

- **Erasmus+ Exchange Semester – University of Łódź** *March 2025 – July 2025*
  Bachelor in Computer Science Łódź, Poland

Language Skills

French: Native Speaker

Arabic: Native Speaker

English: Advanced Knowledge (C1)

German: Intermediate (B1)

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, Haskell, Prolog, HTML/CSS/PHP

Frameworks: Django, FastAPI, React, Nuxt, Next, Vite, Vue

AI: LangGraph, CrewAI, LangChain, RAG, Supabase

Tools: Git, GitHub, Docker, Jupyter Notebook, LangGraph Studio, WordPress, Cursor, Claude Code, Codex

CI/CD: GitHub Actions, Netlify, Vercel, Octopus Deploy